

1-5 B C B. C. D

$$\text{又 } (\beta_1, \beta_2, \beta_3) = (\alpha_1, \alpha_2, \alpha_3) \begin{pmatrix} 1 & 0 & 0 \\ 1 & 1 & 0 \\ 1 & 1 & 1 \end{pmatrix}$$

$$\text{令 } A = \begin{pmatrix} 1 & 0 & 0 \\ 1 & 1 & 0 \\ 1 & 1 & 1 \end{pmatrix}$$

$$R(A) = 3 \Rightarrow A \text{ 可逆}$$

$\therefore \alpha_1, \alpha_2, \alpha_3$ 线性无关

$\therefore \beta_1, \beta_2, \beta_3$ 也线性无关

$$7) -4$$

$$2) 3E$$

$$3) \begin{pmatrix} 0 & -3 \\ 8 & 4 \end{pmatrix}$$

$$4) 9$$

$$5) 4$$

$$\lambda \quad |E-A|$$

$$= \begin{vmatrix} \lambda-1 & -1 & -1 \\ 0 & \lambda-2 & -2 \\ 0 & 0 & \lambda-3 \end{vmatrix} = 0$$

$$\Rightarrow \lambda_1=1 \quad \lambda_2=2 \quad \lambda_3=3$$

$$(E-A)x=0$$

$$(2E-A)x=0$$

$$\begin{pmatrix} 0 & -1 & -1 \\ 0 & 1 & -2 \\ 0 & 0 & 2 \end{pmatrix}$$

$$\begin{pmatrix} 1 & -1 & -1 \\ 0 & 0 & -2 \\ 0 & 0 & 1 \end{pmatrix}$$

$$\alpha_1 = \begin{pmatrix} 1 \\ 0 \\ 0 \end{pmatrix}$$

$$\alpha_2 = \begin{pmatrix} 1 \\ 1 \\ 0 \end{pmatrix}$$

$$(3E-A)x=0$$

$$\hat{P} = \begin{pmatrix} 1 & 1 & \frac{3}{2} \\ 0 & 1 & 2 \\ 0 & 0 & 1 \end{pmatrix}$$

$$\begin{pmatrix} 2 & -1 & -1 \\ 0 & 1 & -2 \\ 0 & 0 & 0 \end{pmatrix} \rightarrow$$

$$P^{-1}AP = \Lambda = \begin{pmatrix} 1 & & \\ & 2 & \\ & & 3 \end{pmatrix}$$

$$Q \begin{pmatrix} 2 & 0 & -3 \\ 0 & 1 & -2 \\ 0 & 0 & 0 \end{pmatrix} \rightarrow$$

$$\begin{pmatrix} 1 & 0 & -\frac{3}{2} \\ 0 & 1 & -2 \\ 0 & 0 & 0 \end{pmatrix}$$

$$\alpha_3 = \begin{pmatrix} \frac{3}{2} \\ 2 \\ 1 \end{pmatrix}$$

$$2 \quad \left(\begin{array}{ccc|ccc} 1 & 1 & 1 & 1 & 0 & 0 \\ 1 & 2 & 1 & 0 & 1 & 0 \\ 1 & 1 & 3 & 0 & 0 & 1 \end{array} \right)$$

$$\left(\begin{array}{ccc|ccc} 1 & 1 & 1 & 1 & 0 & 0 \\ 0 & 1 & 0 & -1 & 1 & 0 \\ 0 & 0 & 2 & -1 & 0 & 1 \end{array} \right) \rightarrow$$

$$\left(\begin{array}{ccc|ccc} 1 & 1 & 0 & \frac{3}{2} & 0 & -\frac{1}{2} \\ 0 & 1 & 0 & -1 & 1 & 0 \\ 0 & 0 & 1 & -\frac{1}{2} & 0 & \frac{1}{2} \end{array} \right) \rightarrow$$

$$\left(\begin{array}{ccc|ccc} 1 & 0 & 0 & \frac{5}{2} & -1 & -\frac{1}{2} \\ 0 & 1 & 0 & -1 & 1 & 0 \\ 0 & 0 & 1 & -\frac{1}{2} & 0 & \frac{1}{2} \end{array} \right)$$

$$\Rightarrow A^{-1} = \begin{pmatrix} \frac{5}{2} & -1 & -\frac{1}{2} \\ -1 & 1 & 0 \\ -\frac{1}{2} & 0 & \frac{1}{2} \end{pmatrix}$$

$$D = \begin{vmatrix} 1 & 2 & 1 & 3 \\ -1 & -1 & -1 & 2 \\ 1 & 1 & 1 & 1 \\ 0 & -1 & 1 & 7 \end{vmatrix}$$

$$= \begin{vmatrix} 1 & 2 & 1 & 3 \\ 0 & 1 & 0 & 5 \\ 0 & -1 & 0 & -2 \\ 0 & -1 & 1 & 7 \end{vmatrix}$$

$$= \begin{vmatrix} 1 & 0 & 5 \\ 0 & 0 & 3 \\ 0 & 1 & 12 \end{vmatrix}$$

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$$= -3$$

$$(1) (1) \begin{pmatrix} 1 & 1 & 1 & 1 \\ 1 & \lambda & 1 & 1 \\ 2 & 2\lambda & \lambda+4 & 1 \end{pmatrix} \rightarrow$$

$$\begin{pmatrix} 1 & 1 & 1 & 1 \\ 0 & \lambda-1 & 0 & 0 \\ 0 & 0 & \lambda+2 & -1 \end{pmatrix}$$

(1) 当 $\lambda \neq 1$ 且 $\lambda \neq -2$ 时有唯一解

(2) 当 $\lambda = -2$ 时, 无解

(3) 当 $\lambda = 1$ 时, 有无穷多解

$$\begin{pmatrix} 1 & 1 & 1 & 1 \\ 0 & 0 & 3 & -3 \\ 0 & 0 & 0 & 0 \end{pmatrix} \rightarrow$$

$$\begin{pmatrix} 1 & 1 & 0 & 2 \\ 0 & 0 & 1 & -1 \\ 0 & 0 & 0 & 0 \end{pmatrix}$$

$$x = k \begin{pmatrix} -1 \\ 1 \\ 0 \\ 0 \end{pmatrix} + \begin{pmatrix} 2 \\ 0 \\ 0 \\ -1 \end{pmatrix}$$

($k \in \mathbb{R}$)

(v)

$$A = \begin{pmatrix} 1 & 1 & 1 & 1 & 1 \\ 3 & 2 & 1 & -3 & X \\ 0 & 1 & 2 & 6 & 3 \\ 5 & 4 & 3 & -1 & y \end{pmatrix} \rightarrow$$

$$\begin{pmatrix} 1 & 1 & 1 & 1 & 1 \\ 0 & -1 & -2 & -6 & X-3 \\ 0 & 1 & 2 & 6 & 3 \\ 0 & -1 & -2 & -6 & y-5 \end{pmatrix} \rightarrow$$

$$\begin{pmatrix} 1 & 1 & 1 & 1 & 1 \\ 0 & 1 & 2 & 6 & 3 \\ 0 & 0 & 0 & 0 & X \\ 0 & 0 & 0 & 0 & y-2 \end{pmatrix}$$

$$\text{rank}(A) = 2$$

$$\Rightarrow \begin{cases} X=0 \\ y=2 \end{cases}$$

t.

$$\begin{pmatrix} 1 & 3 & 2 & 1 & 3 \\ -1 & 1 & 0 & 1 & -1 \\ 1 & 1 & 1 & 0 & 0 \\ -1 & 3 & 1 & 2 & 2 \end{pmatrix} \rightarrow$$

$$\begin{pmatrix} 1 & 3 & 2 & 1 & 3 \\ 0 & 4 & 2 & 2 & 2 \\ 0 & -2 & -1 & -1 & -3 \\ 0 & 4 & 2 & 2 & 2 \end{pmatrix} \rightarrow$$

$$\begin{pmatrix} 1 & 3 & 2 & 1 & 3 \\ 0 & 2 & 1 & 1 & 3 \\ 0 & 2 & 1 & 1 & 1 \\ 0 & 0 & 0 & 0 & 0 \end{pmatrix} \rightarrow$$

$$\begin{pmatrix} 1 & 3 & 2 & 1 & 0 \\ 0 & 2 & 1 & 1 & 1 \\ 0 & 0 & 0 & 0 & 1 \\ 0 & 0 & 0 & 0 & 0 \end{pmatrix} \rightarrow$$

$$\begin{pmatrix} 1 & 0 & \frac{1}{2} & -\frac{1}{2} & 0 \\ 0 & 1 & \frac{1}{2} & \frac{1}{2} & 0 \\ 0 & 0 & 0 & 0 & 1 \\ 0 & 0 & 0 & 0 & 0 \end{pmatrix}$$

秩 $R(A) = 3$

故存在 3 个自由变量 d_1, d_2, d_3

$$\begin{cases} x_3 = \frac{1}{2}d_1 + \frac{1}{2}d_2 \\ x_4 = -\frac{1}{2}d_1 + \frac{1}{2}d_2 \end{cases}$$